Executive summary — English

Question and design. Do options add predictive information for AAPL, AMZN, META, MSFT, NVDA and TSLA? B0 ⊂ B1 ⊂ B2 contains 29/69/138 predictors: price/history; state/surface; flow composition/activity/imbalance. B0 includes multiscale HAR variances, quarticity and HARQ attenuation, signed semivariances, a jump proxy, Parkinson range, returns, volume, SPY/QQQ and intraday calendar terms; it is not a naive benchmark (registration, Corsi 2009, Bollerslev–Patton–Quaedvlieg 2016, Patton–Sheppard 2015). This intraday adaptation compares a winsorized linear model with rank filtering —nominal ridge— and LightGBM using QLIKE, not trading returns. Expanding training uses 60 initial sessions, selection over the last ten and 60-minute purging/embargo; the primary sample contains 419 sessions and 160,832 origins. Forecasts are walk-forward out of sample relative to each fit, with the design fixed after observing the sample. Design.

Answers to the proposal questions. The proposal specified RV30: v3 supports RQ1 in both families, with linear p 0.0439 and tree p 0.0053. RQ2 does not reject at 5%, with p 0.0525 and 0.6631. RQ3 remains partial: asset signs and learning, instrument and expiry diagnostics do not establish all proposed stability. After answering RV30, v4 applied the pre-declared primary RV15/secondary RV5 extension under the short-lived-flow hypothesis, as declared exploration. Proposal · Registration.

Answer and magnitude. Yes within the development sample: B1 > B0 is detected in both families at 30/15 minutes; B2 > B1 > B0 is detected in the linear family at 15/5 under the one-sided H1→H2 sequence at 5%, with RV5 secondary. At 15 minutes, B2/B1 reduces QLIKE by **+0.623%**, with **95% delta CI [0.000335, 0.001932]**, p = 0.0032; at 5, +0.256%. The rescaled CI [+0.184%, +1.061%] uses 100 × delta CI / fixed mean baseline loss, not a bootstrap ratio interval. For trees, B2 ≥ B1 describes positive paired medians only: Holm 0.0870/0.0096. Linear H1 at RV30/RV15 is marginal (0.0439/0.0390), did not reject in v1/v2 and fails the saved bilateral/Holm sensitivity at RV30/RV15; it retains linear H2 at RV15 and tree H1 at RV30/RV15 and linear H1 at RV5, without a complete linear RV15 sequence. Results · Sensitivities.

Horizon comparison. In the primary linear sample, B2/B1 improves more and has positive signs in more assets at 15 minutes (+0.623%, 6/6) than at 5 (+0.256%, 3/6): the observed maximum among the examined horizons is at 15 and does not keep rising as the horizon shortens. This comparison is descriptive; it neither identifies an optimal horizon nor establishes significant differences between horizons. Figures and scope.

Scope and pending evidence. The final 25 sessions and 9,750 origins do not confirm; causal dealer hedging remains unproven. The replication preserves v4: 20 prospective sessions assess primary H1→H2 consistency in linear RV15 and A/B/C with Holm; D pools 25 observed + 20 new sessions; 40 assesses stability without rescue. Approximate H2 power is 14.99/21.62/23.18% at 20/40/45; 45 assumes hypothetical new sessions, not D. The single final look at 335 prospective sessions preserves that sequence and has 80.05% marginal H2 power and 54.98% H1 power, not 80% joint power; no other looks or rescue across looks are allowed. Amendment 3 and assumptions.

Repaired secondary endpoints. Jump AUC is computable and does not reject H1; affine recalibration of LightGBM is arithmetically reproducible but unstable. Neither extends the primary claim. Revision and limits.

Convergence and continuity. RP2, PIT, Phase 8A and RP4 show recurring support for option state, with family/window exceptions; they are not four independent replications or uniformly positive designs, and Gamma/PIT is one evaluation. QLIKE is shared; estimands and windows differ. RP2 block 12 warned about limited power before RP4. RP3 retains its programme sealed on 2026-08-24: required N = 662, a documented zero prior-read counter and estimated reading date 2029-01-30; neither the bank nor the live counter was opened. Table, overlap and dispositions.

Four disclosures.

1. This is the fourth evaluation, without correction for historical search: (a) v1/v2 used bilateral/Holm across four contrasts and v3/v4 a one-sided sequence at 5% per family without cross-family Holm, a less demanding rule registered before v3; (b) B2 added gamma imbalance and indicators after failing in v1/v2; (c) decision 132 fixed empty-window representation after diagnosing v2; (d) primary RV15/secondary RV5, not an independent replication, was pre-declared and sealed locally, without a third-party timestamp; (e) the 2026-08-01 split was fixed on 2026-09-07 with observed data, with the closure rule translated as “If neither rejects, investigation of this mechanism closes with B1>B0 as the main result observed at 30 minutes and B2 not detected at 30/15/5 according to each estimate and interval” and “There is no v5”; the saved bilateral/Holm sensitivity retains linear flow at RV15 and tree state at RV30/RV15 and linear state at RV5, without a complete linear RV15 sequence.

2. The 2026-07-20–2026-08-28 window had been read under another specification and overlaps 20/25 final sessions; the historical README reports mixed/exploratory results, a favorable sensitivity in one of eight B1-inclusive cells and no aggregation change, without independent confirmation.

3. The 120-second PIT cutoff imposes a visibility delay after the record-creation timestamp, rather than a forecast horizon, and remains a proxy that does not establish when a historical client received the record.

4. The provider gap 2025-01-25–2025-02-24 is retained without filling.

Pre-declaration and closure · Chronology, prior readings and limits.

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